

Laboratory II

Optical properties of a massive galaxy cluster

Abstract

The main goal of this exercise is to work with real astronomical data, both spectroscopic and photometric, in order to study the optical properties of a massive galaxy cluster and to identify cluster members. The techniques used include data analysis, filtering and plotting. The results obtained are the calculation of mass and velocity dispersion of the cluster from redshift data, the transmission curves and the spectrum of a brightest cluster galaxy (BCG), a color-magnitude diagram (CMD) with a linear model to the red sequence fitted via 3σ clipping method, the estimation of the projected radius of the cluster and a simplified luminosity function.

1 Introduction

As briefly mentioned in the abstract, the real goal of this exercise behind the several questions and requirements of specific calculations, tables and plots is to get familiar with real astronomical data used by astronomers to study a galaxy cluster, its members and their properties.

The spectroscopic and photometric data employed in this exercise are essential for understanding the cluster's structure and composition. The spectral data provide information about the light emitted by galaxies at different wavelengths, which can be used to determine various physical properties such as redshift, velocity dispersion, and mass of the cluster. In addition, the spectroscopic data can help determine the type of galaxies present in the cluster and identify specific features in their spectra, such as emission or absorption lines.

The spectroscopic data allow us to determine the redshifts of galaxies in the field of view of the cluster, which enables us to identify which galaxies are actually members of the cluster based on their velocities relative to the cluster's mean redshift.

The photometric data, on the other hand, provide information about the brightness and colors of the galaxies in different filters. The cluster under study is MACS 1149, a massive galaxy cluster located at a redshift of approximately $z \approx 0.544$ whose coordinates are about $RA = 177$ deg and $Dec = 22$ deg. Then it is compared with a BCG (at $z \approx 0.396$) template spectrum and transmission curves of the filters used in the observations.

In addition, a color-magnitude diagram (CMD) is required in order to analyze the distribution of galaxies in terms of their colors and magnitudes. This allows us to identify the red sequence of galaxies, which are typically older and more evolved galaxies that have ceased star formation, they had a burst of star formation a long time ago. They also tend to be more massive and are often found in the dense environments of galaxy clusters. By fitting a linear model to the red sequence cluster members can be selected based on their position in the CMD.

Finally, the projected radius of the cluster can be estimated and a simplified luminosity function (LF) can be constructed to study the distribution of galaxy luminosities within the cluster.

For visualisation of the cluster, the DS9 software [1] has been used.

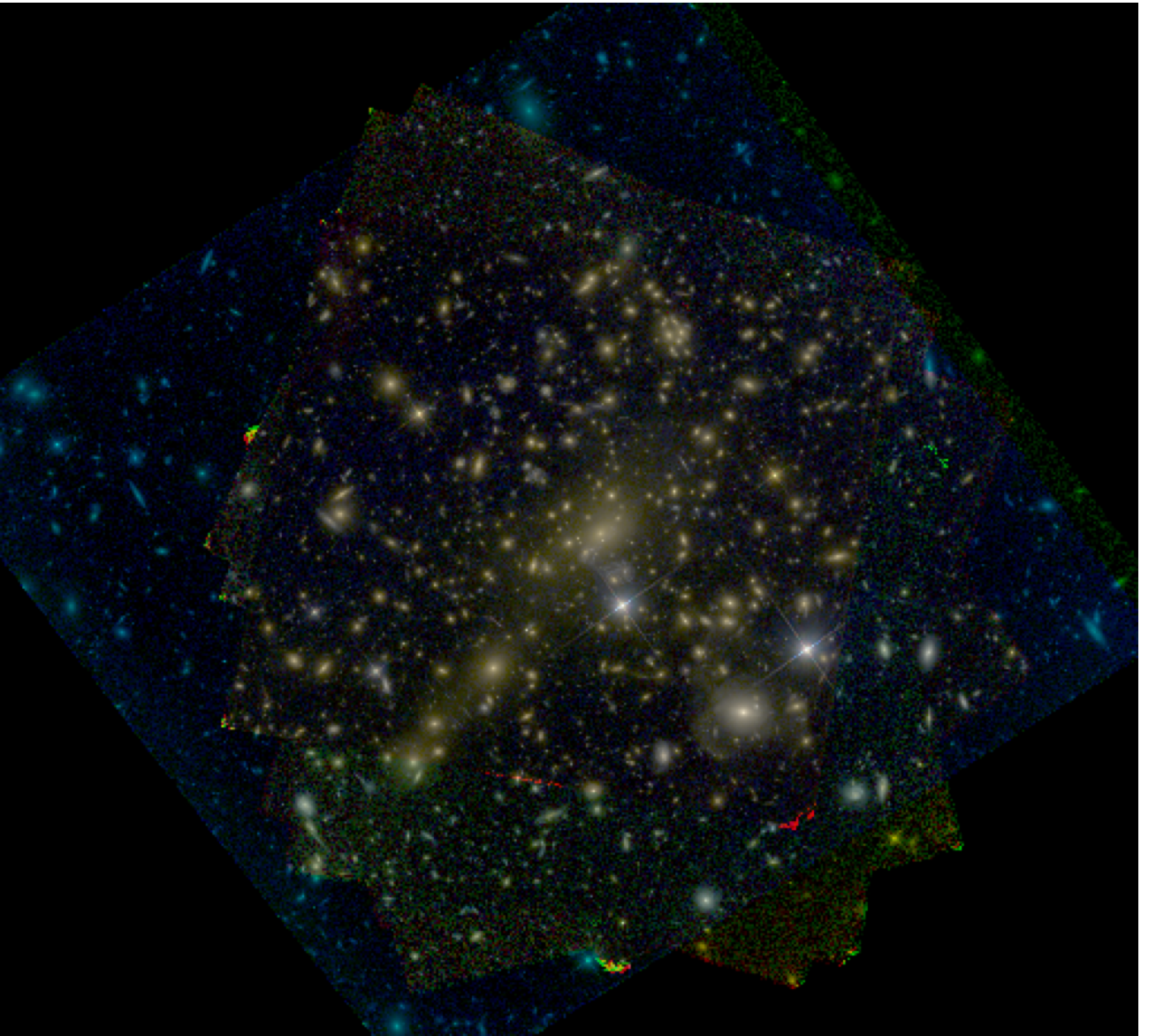


Fig 1. HST colour image of MACS 1149 from DS9.

All the data processing, analysis and calculations are carried out using Python [2] and some of its useful libraries such as SciPy [3], AstroPy [4] and Matplotlib [5], among others.

2 Data

Four datasets (catalogues/files) have been provided for this exercise, which are described below:

- [1] *hlsp_clash_subaru_suprimecam_mac1149_cat.dat*: A BVRcZK photometric catalogue from Subaru Suprime-Cam observations of the MACS 1149 cluster. The data used from this catalogue are the R.A., Dec, the magnitude values (B, V, Rc, Z) and their errors (dB, dV, dRc, dZ).
- [2] *M1149_v2.0_zcat_MUSE_GLASS.dat*: A spectroscopic redshift catalogue from MUSE and GLASS observations of the MACS 1149 cluster. The data used from this catalogue are the R.A., Dec and redshift (z).
- [3] *BCG_spectrum.dat*: A 1D spectrum of the brightest cluster galaxy (BCG) in a galaxy cluster at $z = 0.396$ observed with VLT/MUSE. The wavelength units are in Angstroms ($\text{\AA}$) and the Flux is in arbitrary units.
- [4] *macs1149_hst_rgb_all.fits*: HST colour image of MACS 1149 (R: F140W, F160W; G: F105W, F125W; B: F606W, F814W), which can be managed with software like DS9 [1] or similar for visualisation purposes, it is not used for calculations in this exercise.

During the Laboratory it is requested to create two more catalogues/datasets from the original ones, which are the following:

- [MYFILE1]: A catalogue of galaxies in the cluster field selected by their spectroscopic redshift within the range $0.5272 \leq z \leq 0.5660$ which can be considered cluster members. This is the first step towards the filtering of the cluster members.
- [MYFILE2]: A catalogue of the spectroscopically- and photometrically-selected cluster members which includes the relevant information.

The instruments used to gather the data are the following:

- **Subaru/Suprime-Cam**: A wide-field optical camera mounted on the Subaru Telescope, located at the Mauna Kea Observatory in Hawaii. It is designed for deep imaging and surveys of large areas of the sky.
- **VLT/MUSE**: The Multi Unit Spectroscopic Explorer (MUSE) is an integral field spectrograph mounted on the Very Large Telescope (VLT) at the Paranal Observatory in Chile. It provides spatially resolved spectroscopy over a wide field of view, allowing detailed studies of galaxies and galaxy clusters.
- **HST**: The Hubble Space Telescope (HST) is a space observatory that has provided high-resolution imaging and spectroscopy across a wide range of wavelengths for over three decades.

3 Answer to questions

Question 1

The first requirement is to create [MYFILE1], a catalogue of galaxies within the redshift range $0.5272 \leq z \leq 0.5660$. A histogram of the spectroscopic redshift distribution of these galaxies is created and a Gaussian distribution is fitted to it. The mean and the standard deviation are provided in the legend of Fig. 1a.

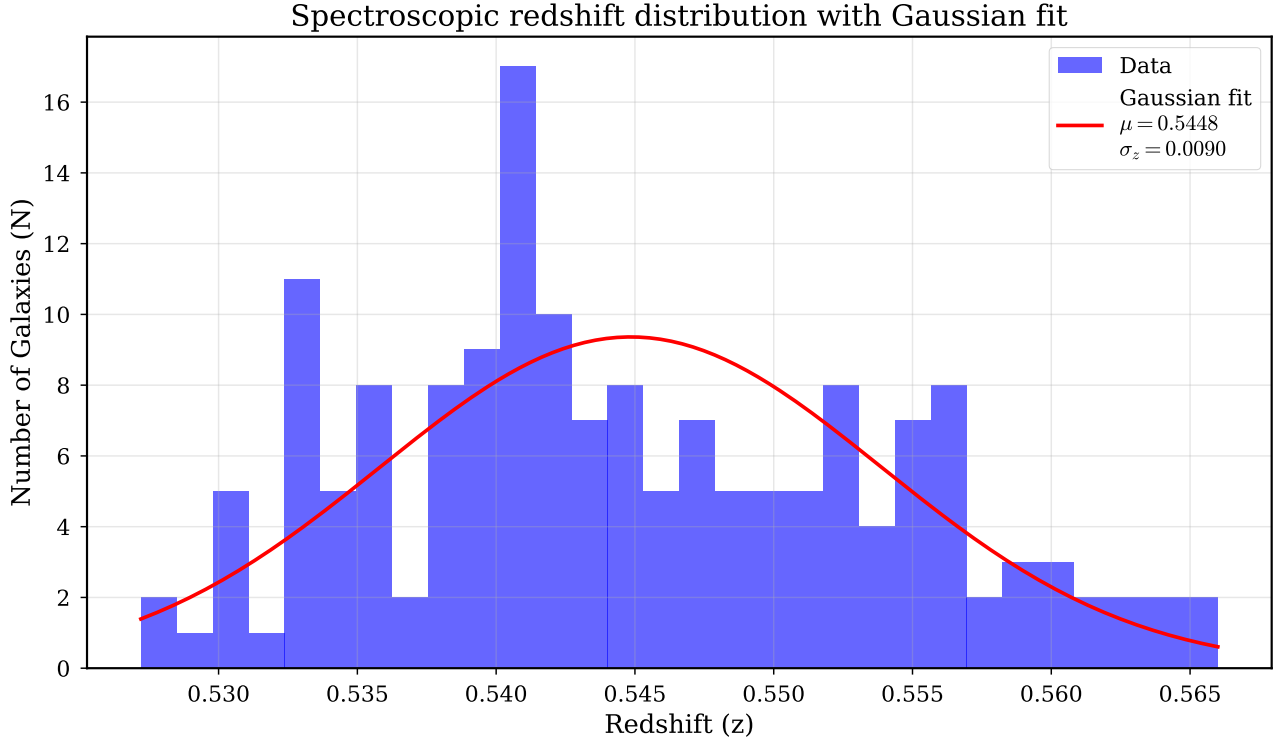


Fig 1a. Spectroscopic redshift distribution of galaxies within the cluster field (blue histogram) with a Gaussian fit (red curve). The mean, μ , represents the mean redshift of the cluster, z_{cl} , while the standard deviation indicates the spread in redshift values of the cluster members, σ_z .

From the spectroscopic redshift the velocity dispersion, σ_v , of the cluster is calculated using the following equation:

$$\sigma_v = c \cdot \frac{\sigma_z}{1 + z_{cl}} \quad (1)$$

where c is the speed of light. The result of the velocity dispersion is the following:

$$\sigma_v \approx 1755.03 \text{ } \epsilon \text{ } [-3427.05, 4107.68] \text{ km/s,}$$

and the interval of velocities shown is the corresponding to the redshift range, which has been calculated using the same Eq. (1) for the minimum and maximum redshift values of the range. For the velocity distribution of these galaxies with respect to the mean cluster redshift obtained above, a histogram (Fig. 1b) can also be plotted and a Gaussian distribution can be fitted to it. Again, the mean and standard deviation of the Gaussian distribution on the legend of Fig. 1b.

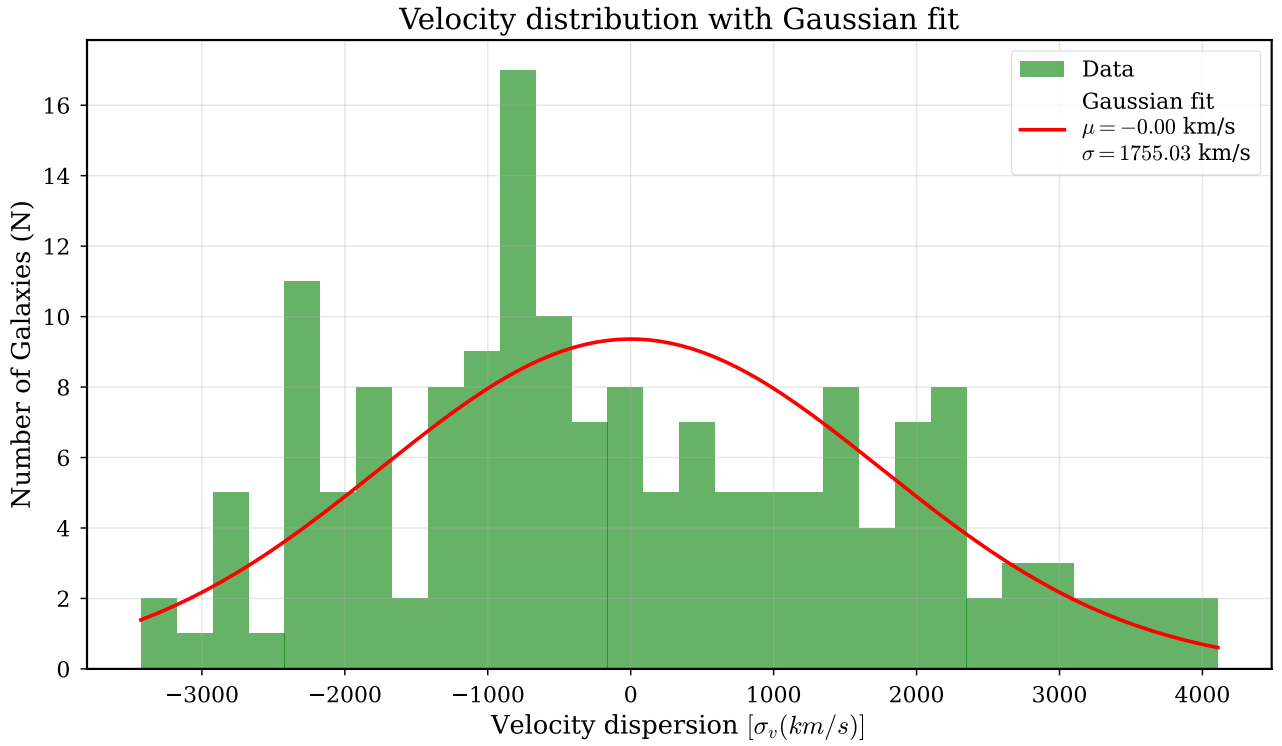


Fig 1b. Velocity distribution of galaxies within the cluster field (green histogram) with a Gaussian fit (red curve). The mean, μ , represents the mean velocity of the cluster, v_{cl} , and the standard deviation, σ , indicates the spread in velocity values of the cluster members, σ_v .

It can be appreciated that both histograms (Figs. 1a and 1b) look very similar, as they should, since the velocity distribution is derived from the redshift distribution by the Eq. (1). The mean values are consistent with each other, confirming the reliability of the measurements. The standard deviation in redshift translates to a corresponding standard deviation in velocity, which reflects the same underlying physical properties of the cluster. The small variations of these two values are due to rounding errors.

The velocity dispersion obtained, $\sigma_v \approx 1755.03$ km/s, is quite high compared to typical values for relaxed galaxy clusters, which usually range from a few hundred to around 1000 km/s. According to the literature [6, 7, 8] the MACS 1149 is a merging cluster, which can explain the high velocity dispersion value calculated. The two subclusters may be relaxed individually, but the whole cluster is not fully relaxed, and the high velocity dispersion reflects the complex dynamics of the merging process. Regardless, the Gaussian fit to the velocity distribution is reasonable.

Finally, the total mass of MACS 1149 within a radius of 3 Mpc is estimated using the virial theorem, which can be expressed as a relation between the mass of the cluster, velocity dispersion, and radius:

$$M_{dyn} = \frac{3\pi R_G \sigma_v^2}{2G} \quad (2)$$

$$\frac{M_{dyn}}{M_\odot} \approx 1.1 \times 10^{15} \left(\frac{\sigma_v}{1000 \text{ km/s}} \right)^2 \left(\frac{R_G}{1 \text{ Mpc}} \right)$$

where M_{dyn} is the total mass of the cluster, σ_v is the velocity dispersion, R is the radius within which the mass is being estimated (3 Mpc in this case), and G is the gravitational constant. The result of the mass estimation is the following:

$$M_{MACS1149} \approx 1.02 \times 10^{16} M_{\odot}$$

where $M_{\odot}$ is the solar mass. This estimation is overestimated due to the high velocity dispersion calculated previously, which is a consequence of the merging nature of the cluster. This is due to the fact that the virial theorem assumes a relaxed system in equilibrium, which is only partly true for MACS 1149 as discussed before.

Question 2

The BVRcZ transmission curves obtained from the Subaru/SuprimeCam [9] and the BCG spectrum (file [3], $z = 0.396$), redshifted to the MACS 1149 redshift ($z = 0.544$) are plotted in Fig. 2.

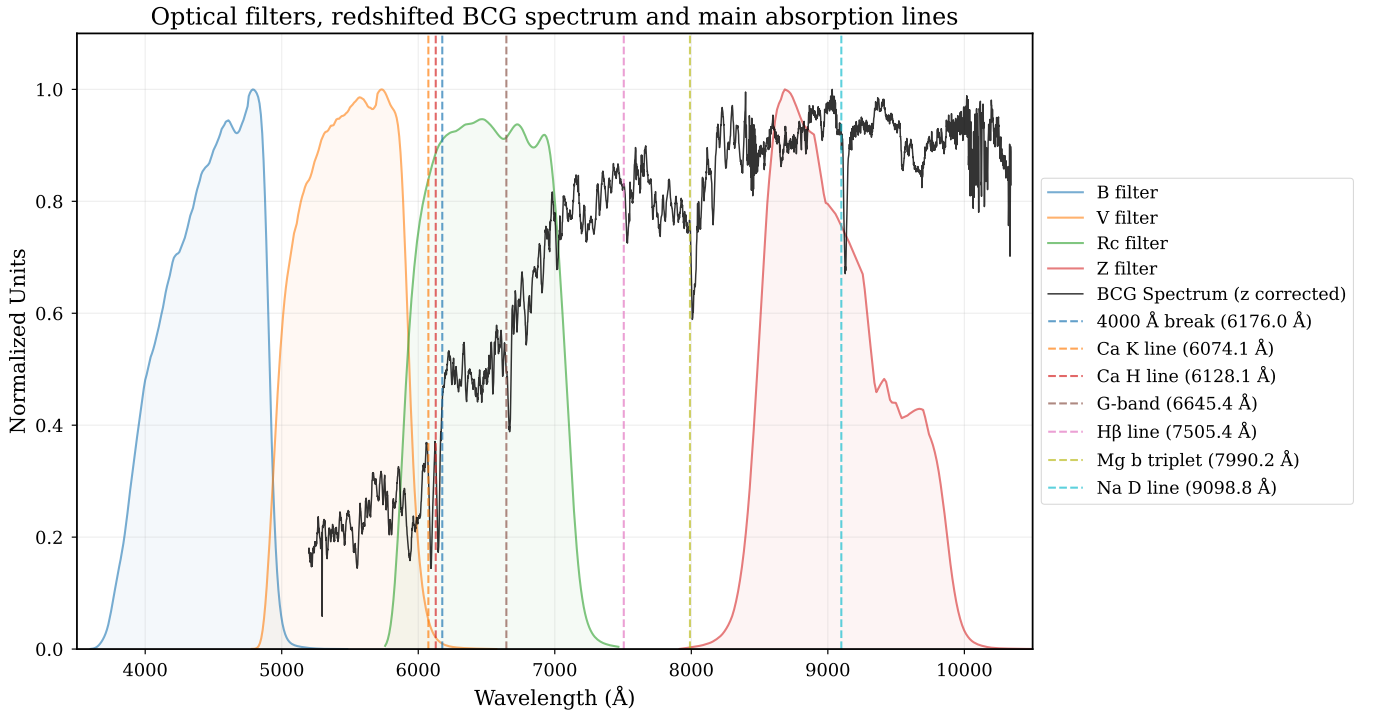


Fig 2. The BVRcZ transmission curves (colored areas) and the BCG spectrum (black line), redshifted to the MACS 1149 redshift of $z = 0.544$. The main absorption lines that are characteristics of the early-type galaxy population are marked with dashed vertical lines.

To redshift the BCG spectrum to the MACS 1149 redshift, the following factor is used:

$$f_z = \frac{1 + z_{MACS1149}}{1 + z_{BCG}} = \frac{1 + 0.544}{1 + 0.396} \approx 1.106 \quad (3)$$

where $z_{MACS1149} = 0.544$ is the redshift of the cluster and $z_{BCG} = 0.396$ is the redshift of the BCG spectrum.

The most effective filters for observing the BCG at the MACS 1149 redshift are the Rc (green) and Z (red) filters because the 4000 Å break and some other absorption lines characteristic of early-type galaxies fall within these filters' transmission curves, as can be seen in Fig. 2.

Question 3

A match between the spectroscopic catalogue (file [1]) and the photometric catalogue [MYFILE1] is performed with TOPCAT [10]. A colour-magnitude diagram (CMD) is plotted with a linear model fitted to the red sequence via 3σ clipping method of the spectroscopically-confirmed cluster galaxies and a 1σ region around the fitted line is marked in Fig. 3a.

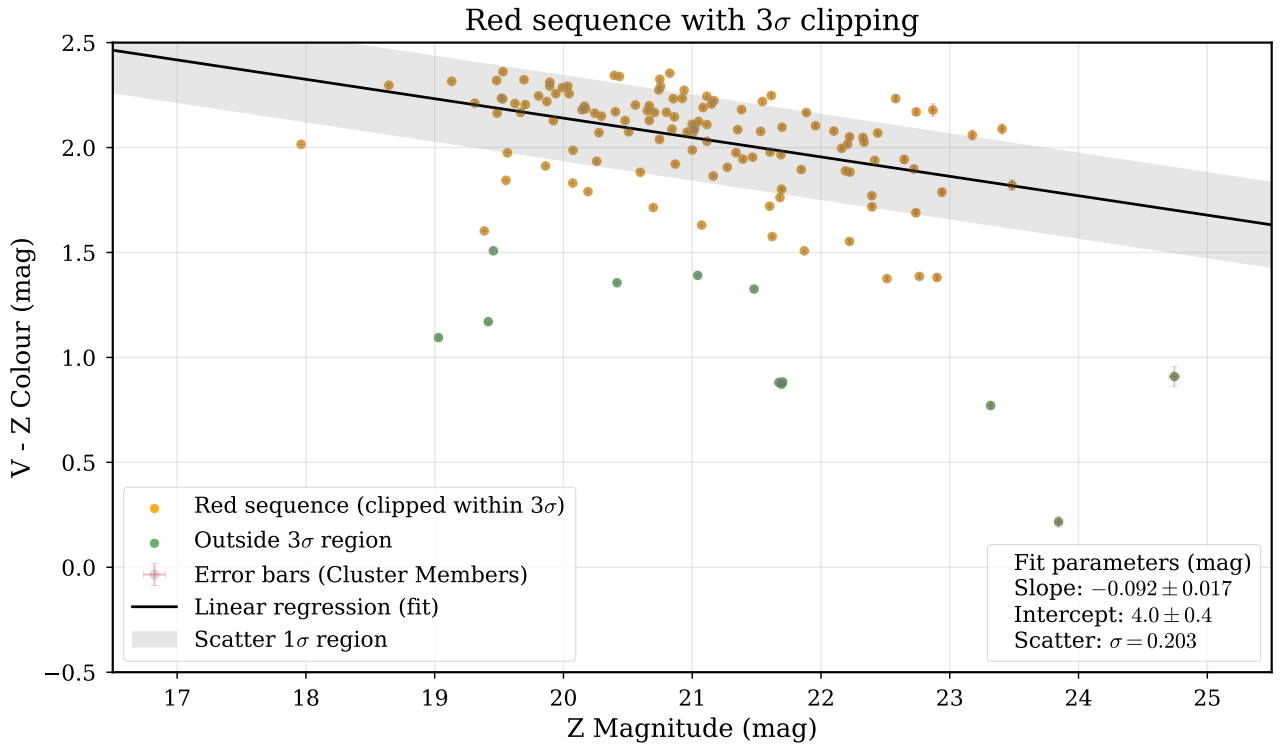


Fig 3a. Colour-magnitude diagram (CMD) of the spectroscopically-confirmed cluster galaxies. The black line represents the linear model fitted to the red sequence via 3σ clipping method, with a slope of $a = -0.092 \pm 0.017$, an intercept of $b = 4.0 \pm 0.4$ and a scatter of $\sigma = 0.203$ mag. The shaded area indicates the 1σ region around the fitted line.

Most of the spectroscopically-confirmed cluster galaxies fall within the 1σ region around the fitted line, which suggests that the linear model is a good representation of the red sequence of the cluster. Those outside this region could be blue cluster members or foreground/background galaxies, and the furthest ones are ruled out by the 3σ clipping method. The tilt of the red sequence is negative, indicating that brighter galaxies tend to be redder, which is consistent because more massive galaxies have older stellar populations and higher metallicities, leading to redder colors.

Another CMD is asked with all the sources for which the photometry is available in the field with $Rc \leq 24$ (file [1]) and 89 of the total 132 spectroscopically-confirmed cluster members that fall within 1σ of the fitted line are differentiated.

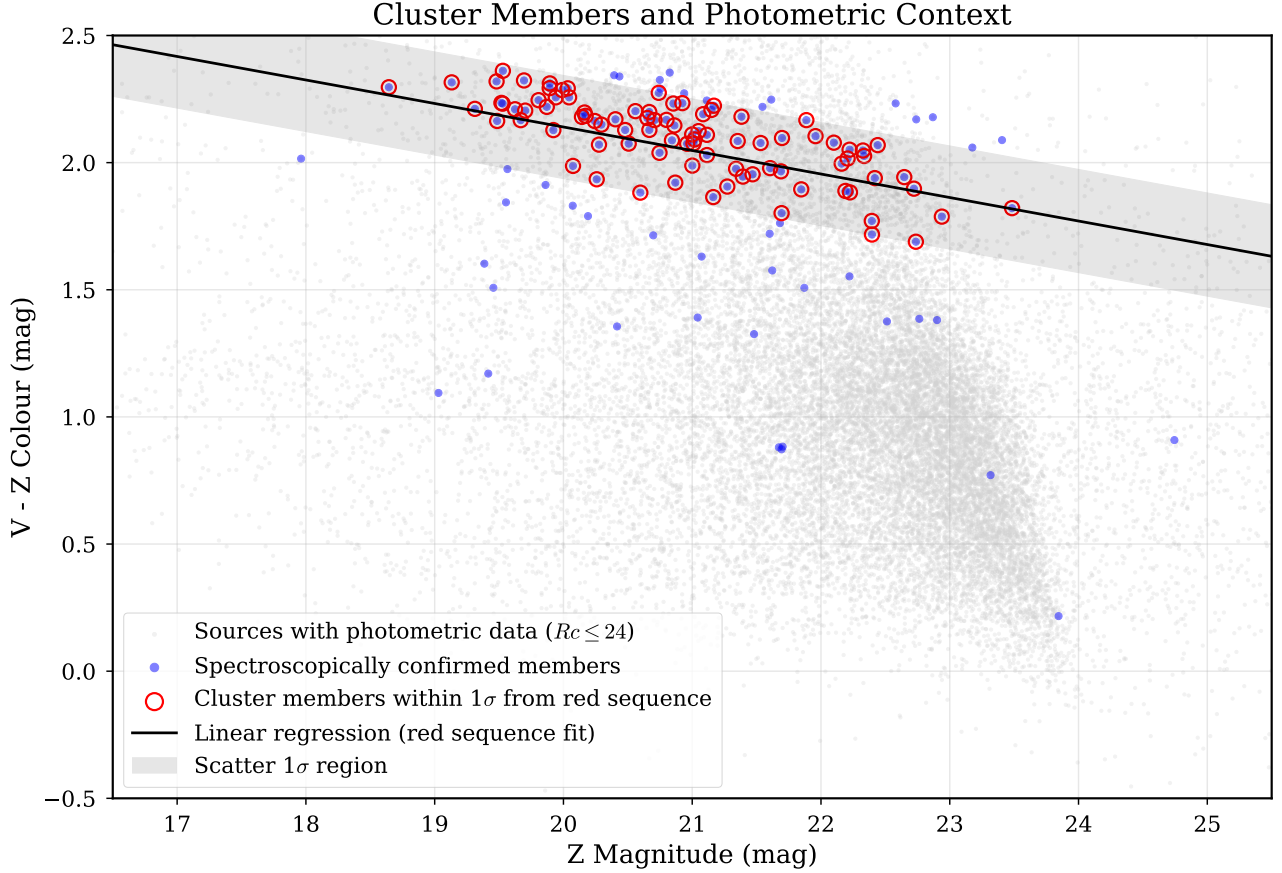


Fig 3b. Colour-magnitude diagram (CMD) of the spectroscopically-confirmed cluster galaxies. The black line represents the linear model fitted to the red sequence. The red circles highlight the cluster members that fall within 1σ of the fitted line, which are under the shaded area. The grey points represent all the sources for which the photometry is available in the field with $Rc \leq 24$. The slope, intercept and scatter of the fitted line are the same as in Fig. 3a.

The galaxies detected photometrically in the field with $Rc \leq 24$ (grey points) and that fall within 1σ of the fitted line (shaded area) are considered photometrically-selected cluster member candidates. These candidates are highlighted with red circles in Fig. 3b. These galaxies are likely to be cluster members because they lie on the red sequence and are close to some spectroscopically confirmed cluster members.

Finally, the [MYFILE2] catalogue of spectroscopically- and photometrically-selected cluster members is created with the relevant information. It also includes the spectroscopic blue galaxies that have been removed from the fit to perform the 3σ clipping method. This catalogue is composed of a total of 2073 galaxies.

Question 4

The projected radius of the cluster is estimated using the angular separation between the BCG and the farthest cluster member identified in [MYFILE2]. The angular separation is calculated using the Astropy [4] library in Python [2]. The Cosmology Calculator from Ned Wright [11] provided in the Laboratory I guide has also been used to verify the results.

To calculate the projected radius it is necessary to know the angular diameter distance, D_A , at the cluster redshift of $z = 0.5448$. Using the Flat Lambda Cold Dark Matter (FlatLambdaCDM) cosmological model with parameters $H_0 = 70 \text{ km/s/Mpc}$ and $\Omega_m = 0.3$, and the result is:

$$D_A(z = 0.5448) \approx 1316 \pm 11 \text{ Mpc},$$

which is calculated using the `.angular_diameter_distance()` function from Astropy. Using Ned Wright's Cosmology Calculator [11] the same result is obtained:

$$D_A(z = 0.5448) \approx 1316.2 \text{ Mpc}.$$

The maximum angular separation found between the BCG and the farthest cluster member is calculated with the `.separation()` function (from Astropy as well), and the result is:

$$\theta_{max} \approx 0.005882 \text{ radians} \approx 20.22 \text{ arcmin}.$$

Finally, the projected radius, R , is calculated using the following equation:

$$R = D_A \cdot \theta_{max} \tag{4}$$

and the result is:

$$R \approx 7.74 \pm 0.06 \text{ Mpc},$$

which is big compared to the initial radius of 3 Mpc used in Question 1 for the mass estimation of the cluster. This large radius is due to the fact that photometrically-selected cluster members are included in [MYFILE2], which can be located farther from the cluster center than the spectroscopically-confirmed members alone. If the projected radius is calculated only with the spectroscopically-confirmed cluster members, the results are the following:

$$\theta_{max} \approx 0.000456 \text{ radians} \approx 1.55 \text{ arcmin}$$

for the maximum angular separation and

$$R \approx 0.594 \pm 0.008 \text{ Mpc}$$

for the projected radius, which is more consistent with the initial radius used for the mass estimation and it indicates that the spectroscopically confirmed cluster members are located in the central region of the cluster.

The next step is to calculate the absolute magnitude, M_{Rc} , and luminosity, L_{Rc} (in units of $L_{\odot}$) in the Rc band for all the selected cluster members in [MYFILE2]. The absolute magnitude is calculated using the following equation:

$$M_{Rc} = m_{Rc} - \mu = m_{Rc} - 5 \log_{10}(D_L/10\text{pc}), \quad (5)$$

where m_{Rc} is the apparent magnitude in the Rc band, μ is the distance modulus (without the K-correction, which is neglected here for simplicity), and D_L is the luminosity distance to the cluster, which is calculated with the `.luminosity_distance()` function from Astropy and it is approximately:

$$D_L(z = 0.5448) \approx 3140 \pm 60 \text{ Mpc}.$$

Again, the same result is obtained using Ned Wright's Cosmology Calculator [11]:

$$D_L(z = 0.5448) \approx 3141.2 \text{ Mpc}.$$

The luminosity in the Rc band is calculated using the following equation:

$$L_{Rc} = 10^{(M_{\odot,Rc} - M_{Rc})/2.5} \quad (6)$$

where $M_{\odot,Rc} = 4.60$ mag is the absolute magnitude of the Sun in the Rc band according to Willmer [12].

The 10 brightest galaxies in Rc band are selected from [MYFILE2] and their measurements (R.A., Dec, Rc , M_{Rc} , L_{Rc}) are included in the following table:

R.A. (deg)	Dec (deg)	Rc (mag)	M_{Rc} (mag)	L_{Rc} ($10^{11} L_{\odot}$)
177.219132	22.321810	17.31	-25.18	8.21
177.324042	22.319548	17.49	-25.00	6.98
177.387736	22.522645	17.56	-24.92	6.49
177.161159	22.289658	17.57	-24.91	6.43
177.644176	22.591469	17.80	-24.68	5.21
177.451894	22.483447	17.81	-24.67	5.17
177.571541	22.274984	18.02	-24.47	4.27
177.193226	22.631538	18.33	-24.15	3.20
177.582956	22.300010	18.42	-24.06	2.94
177.420659	22.136813	18.47	-24.01	2.82

Tab 1. Top 10 brightest cluster members in Rc band from [MYFILE2] with their R.A., Dec, apparent magnitude (Rc), absolute magnitude (M_{Rc}) and luminosity (L_{Rc}).

These brightest galaxies are likely to be massive elliptical galaxies, which are common in the dense environments near the center of galaxy clusters. The high magnitudes and luminosities indicate that these galaxies contain more stars and have higher stellar masses than other galaxies.

The last request in this question is to compute the median Rc luminosity of the cluster member catalogue, $\overline{L_{Rc}}$, and the total Rc luminosity of the galaxy cluster within the radius R, $L_{Rc,total}$. Then, based on the results from Question 1, the mass-to-light ratio, $(M_{\odot}/L_{\odot})_{Rc}$, of MACS 1149 is estimated.

These parameters are initially calculated using all the cluster members in [MYFILE2], which correspond to a projected radius of $R \approx 7.74$ Mpc. However, due to the ambiguity of the question two more cases are considered: 0.594 Mpc, which would create an alternative [MYFILE2] including only spectroscopically-confirmed members, and 3 Mpc, the projected radius used for mass estimation in Question 1. All these results are presented in the next table:

R_G (Mpc)	$\overline{L_{Rc}}(\times 10^{11} L_{\odot})$	$L_{Rc,total}(\times 10^{11} L_{\odot})$	$(M_{\odot}/L_{\odot})_{Rc}$
7.74	0.064 ± 0.012	330 ± 20	310 ± 60
0.594	0.100 ± 0.020	22.0 ± 2.5	4600 ± 1000
3	0.063 ± 0.019	120 ± 12	840 ± 170

Tab 2. Median luminosity, $\overline{L_{Rc}}$, total luminosity, $L_{Rc,total}$, and mass-to-light ratio, $(M_{\odot}/L_{\odot})_{Rc}$, in Rc band for MACS 1149 for different radii. The errors provided for the median luminosity, total luminosity and mass-to-light ratio have been calculated using quadratic error propagation.

The mass used to calculate the mass-to-light ratio in all cases is the same as in Question 1, which is $M_{MACS1149} \approx 1.02 \times 10^{16} M_{\odot}$. Then, for the calculated projected radius ($R = 7.74$ Mpc, which is believed to be the one asked in the question), the mass-to-light ratio is a bit higher than typical values found in the literature for massive galaxy clusters. A high mass-to-light ratio as the present indicates that the cluster contains a significant amount of dark matter, which contributes to the total mass but does not emit light.

For the initial projected radius used for the mass estimation ($R = 3$ Mpc) the mass-to-light ratio is still high, but more reasonable. Finally, for the smallest radii considered ($R = 0.594$ Mpc) the mass-to-light ratio is extremely high because a big portion of the luminosity of the cluster is not being taken into account.

In this case it is hard to get any conclusion about the dark matter content of the cluster due to the uncertainties in the mass and luminosity estimations discussed. Nevertheless, for the calculated projected radius, $R = 7.74$ Mpc, the most likely explanation is that both the mass and the projected radius are overestimated, then the two effects compensate each other and the mass-to-light ratio ends up having the expected order of magnitude.

Question 5

A simplified luminosity function (LF) is constructed using the Rc absolute magnitudes of the cluster members in [MYFILE2], M_{Rc} . The cluster members from the mentioned file are filtered

according to the lab instructions: only galaxies with an apparent magnitude value $R_c \leq 24$ are considered and the BCG is removed. The LF is plotted as $\log(N(< M_{R_c}))$ versus M_{R_c} , where $N(< M_{R_c})$ is the number of galaxies brighter than absolute magnitude M_{R_c} in the following figure:

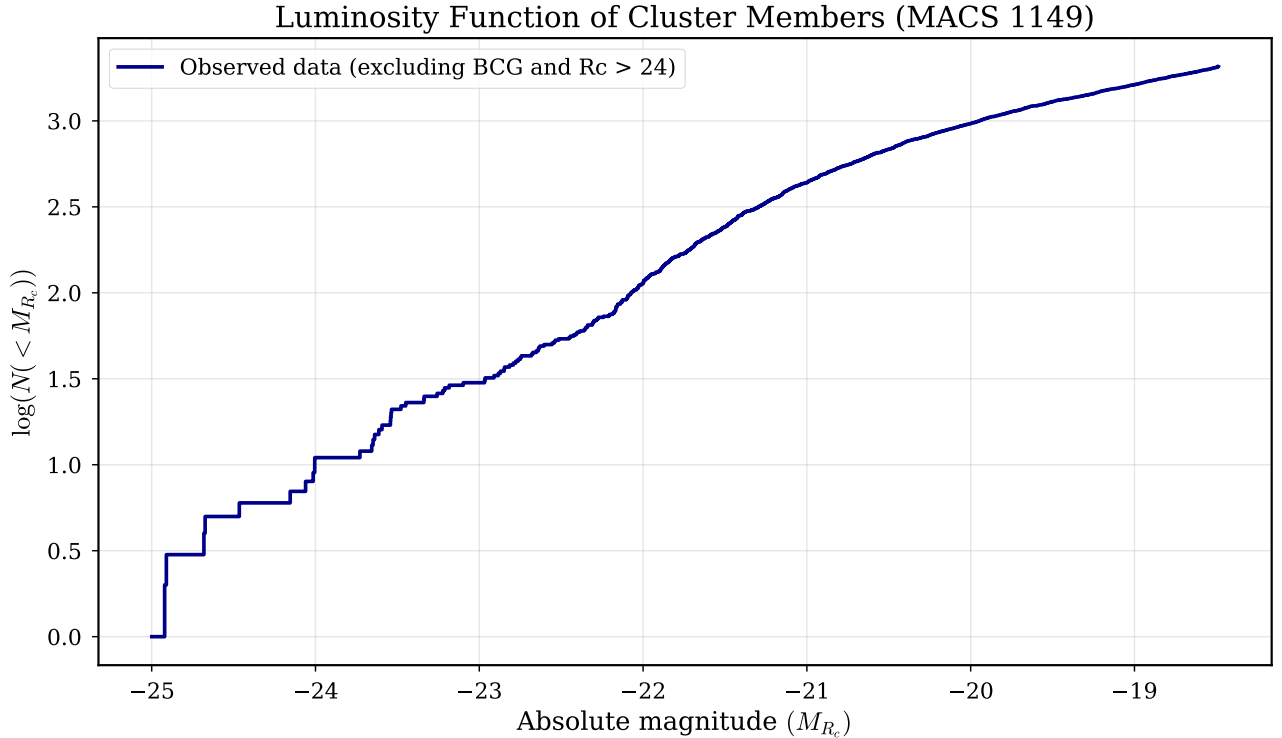


Fig 4. Simplified luminosity function (LF) of the cluster members in R_c band. The BCG has been removed and only galaxies with $R_c \leq 24$ are considered.

The shape of the LF indicates that on one hand, there are few very bright galaxies, hence the LF starts at low values of N . On the other hand, there are many faint galaxies, hence N rises quickly towards higher magnitudes. This behavior is consistent with the red sequence, which determines that the majority of cluster members are faint, red galaxies. However, the brightest cluster members, which are few in number, are located at the bright end of the LF.

4 Conclusions

The redshift distribution and velocity dispersion of the cluster members have been plotted in Figs. 1a and 1b. The high velocity dispersion obtained suggests a possible merging nature of MACS 1149, which means that it is not a relaxed system, in agreement with the literature [6, 7, 8].

This implies that the virial theorem used for mass estimation within a radius of 3 Mpc may not be fully applicable, and the calculated mass of approximately $1.02 \times 10^{16} M_{\odot}$ is likely overestimated.

The BVRcZ transmission curves and the BCG spectrum redshifted to the MACS 1149 redshift have been plotted in Fig. 2. The Rc and Z filters have been identified as the most effective for

observing the BCG at the cluster redshift as the 4000Å break falls within these bands.

A colour-magnitude diagram (CMD) of the spectroscopically-confirmed cluster galaxies has been created, with a linear model fitted to the red sequence via 3σ clipping method (Fig. 3a). A second CMD including all sources with photometry available in the field and $R_c < 24$ has also been plotted, highlighting the spectroscopically-confirmed cluster members that fall within 1σ of the fitted line (Fig. 3b).

The projected radius of the cluster has been estimated to be 7.74 ± 0.06 Mpc, based on the angular separation between the BCG and the farthest cluster member. The absolute magnitudes and luminosities in R_c band for all the selected cluster members have been calculated and listed in Table 1 for the 10 brightest galaxies. The median R_c luminosity, $\bar{L}_{Rc}$, total R_c luminosity, L_{Rc} , and mass-to-light ratio, $(M_\odot/L_\odot)_{Rc}$, of MACS 1149 in R_c band have been computed for different radii (Table 2). The mass-to-light ratio for the calculated projected radius is slightly higher than typical values, indicating a significant amount of dark matter in the cluster.

Finally, a simplified luminosity function (LF) of the cluster members in R_c band has been plotted in Fig. 4. The LF shape is consistent with the red sequence expectations, showing few bright galaxies and many faint galaxies.

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